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# The design cycle, end to end

Lecture 1 · CADE30008 Flight Dynamics & Advanced Control

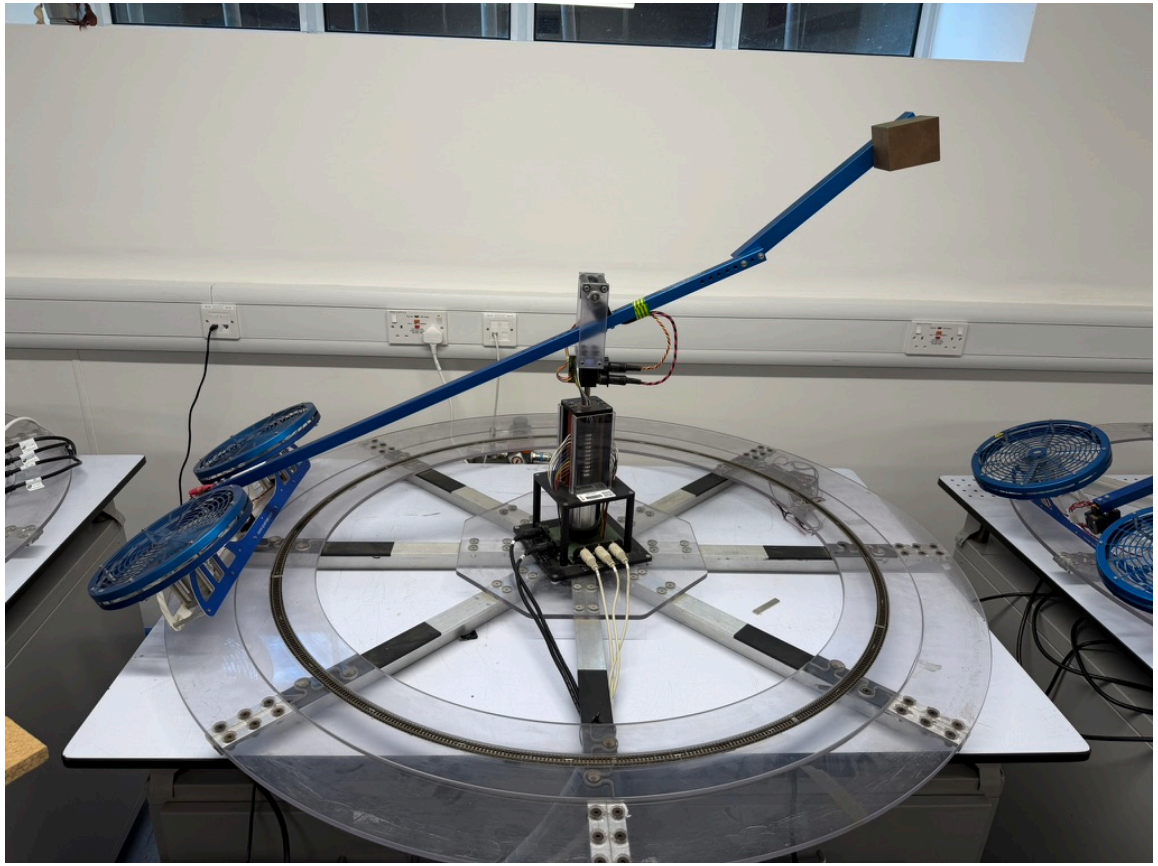
Dr. Steve Bullock

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# The claim

Made now, tested in half an hour

# This machine cannot be flown by hand



**One of you is going to try.**

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# Why control exists

And the design cycle our methods are part of

# The mathematics this unit starts from

- Laplace transforms, transfer functions, poles and zeros
- Standard second-order form: what $\zeta$ and $\omega_n$ do to a response
- Bode plots from first- and second-order factors
- PID, in its parallel form

**If any of that feels distant: the prerequisite diagnostic. Eleven questions, worked feedback on every option, no marks.**

# Control is a cycle

1. **Understand the plant.** How it performs, ideally not what the datasheet says
2. **State the requirement.** What 'good' looks like, in numbers, before designing
3. **Design.** Structure, gains and other parameters, a prediction
4. **Validate in simulation.** Does it meet the requirement?
5. **Test on hardware.** Does it *really*?
6. **Go round again**, because 5 disagreed with 4

> **Today you do all six. Quickly. Once.** To put the rest of the course in context: everything after today is one of these steps, done properly.

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# Feedback

## What it gives, and what it costs

# Feedback is a trade, not a fix

**You get, for free:**

- disturbance rejection
- insensitivity to the plant
- the ability to stabilise something unstable

**You pay, unavoidably:**

- sensor noise reaches the output
- the loop can go unstable, which the open plant could not
- **more gain brings more of both**

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# The machine

Three axes, two motors

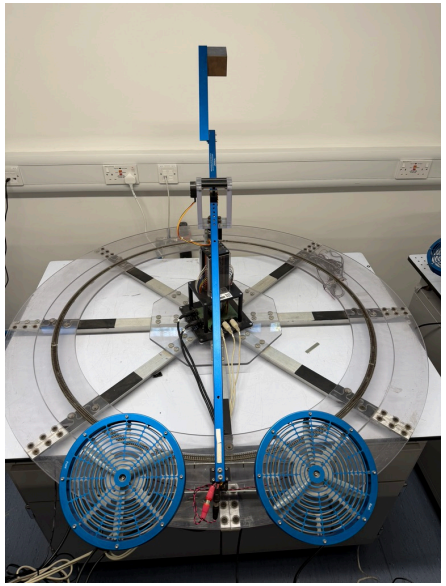
# It is a tandem-rotor helicopter






Quanser model it on the **Boeing HC-1B Chinook**. Or, if helicopters are not your thing: **half a quadcopter**.

# Elevation, pitch, travel



| | | |
|---|---|---|
| **Elevation** ε | height of the beam | zero when level |
| **Pitch** ρ | tilt of the motor pair | front motor higher |
| **Travel** λ | swing about the column | counter-clockwise from above |

# Before anything is switched on




**The circled rocker switch is the emergency stop.** No separate button, no software interlock. Know where it is before the motors spin.

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# The attempt

Two joysticks, three axes

YOU DO

# Fly it

**Open loop. Two joysticks. All three axes.**

Take off and fly for 10 seconds. It ends when you hit an axis limit.

Everyone else: count.

# Why a person loses to this

- The oscillation **does not die away** on its own
- Every correction arrives **slightly too late**, and makes it bigger
- Fixing elevation with pitch **sends it round the track**
- You are controlling **three coupled things with two levers**

**None of this is about reflexes.**

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# One axis

## And a model of it

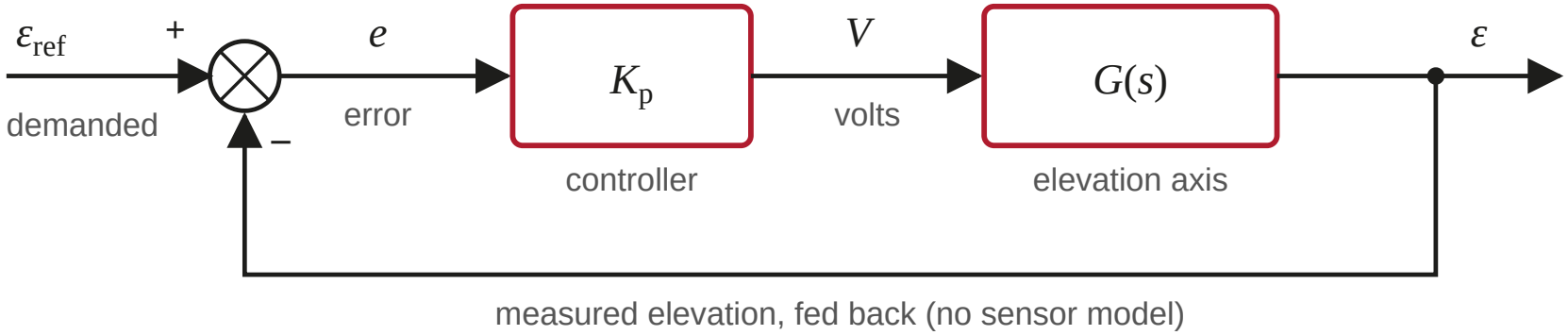
# Elevation, measured

$$G(s) = \frac{\varepsilon(s)}{V(s)} = \frac{K\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}s + \omega_{\mathrm{n}}^2}$$

> K ≈ **3.4** deg/V · $\omega_n$ ≈ **1.0** rad/s · ζ ≈ **0.06**

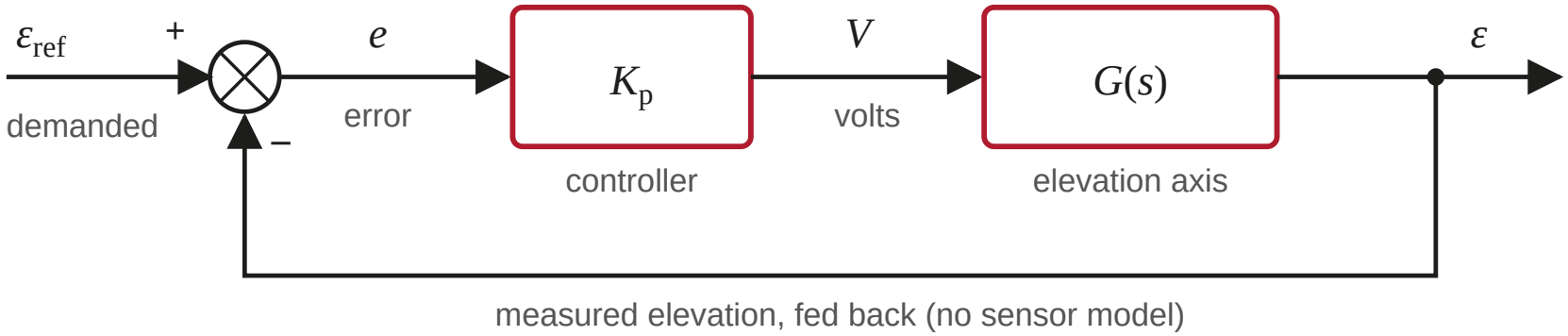
- **Stable.** Disturb it and it does come back
- ζ = 0.06 is **almost no damping**: six-second period, about a minute to settle
- Nudge it and walk away, and it is still moving when you return

# So close a loop around it




Measure the elevation, compare it with the one you asked for, and drive the motors depending on the difference.

# Reminder: closed-loop TF & characteristic polynomial




$$e = \varepsilon_{\text{ref}} - \varepsilon, \qquad V = K_p e, \qquad \varepsilon = G(s)\, V$$

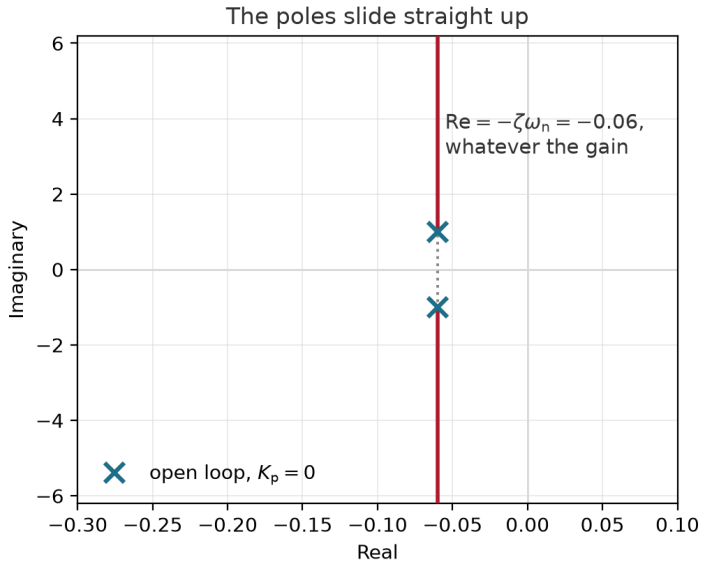
$$\varepsilon\,(1 + K_p G) = K_p G\, \varepsilon_{\text{ref}} \qquad \Longrightarrow \qquad \frac{\varepsilon}{\varepsilon_{\text{ref}}} = \frac{K_p G}{1 + K_p G}$$

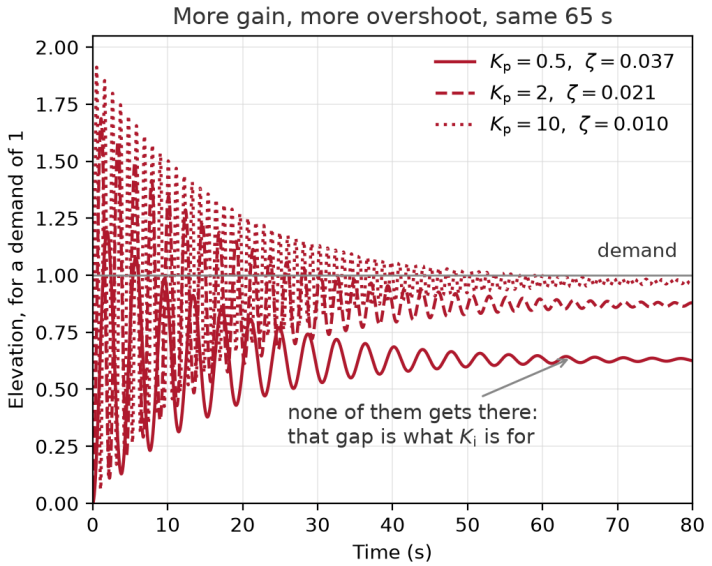
$$G = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \implies \text{denominator} = s^2 + 2\zeta\omega_n s + \omega_n^2\,(1 + K K_p)$$

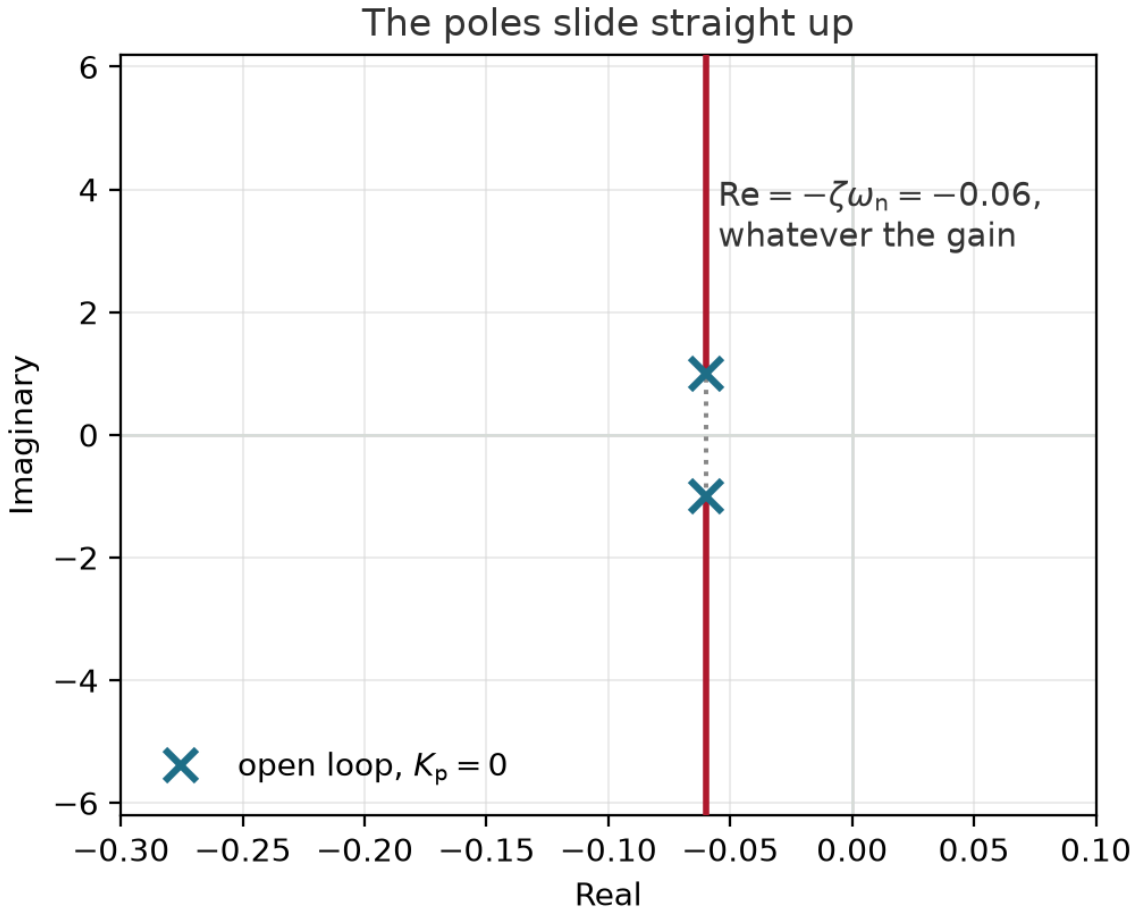
# "Turn the gain up" does not work here

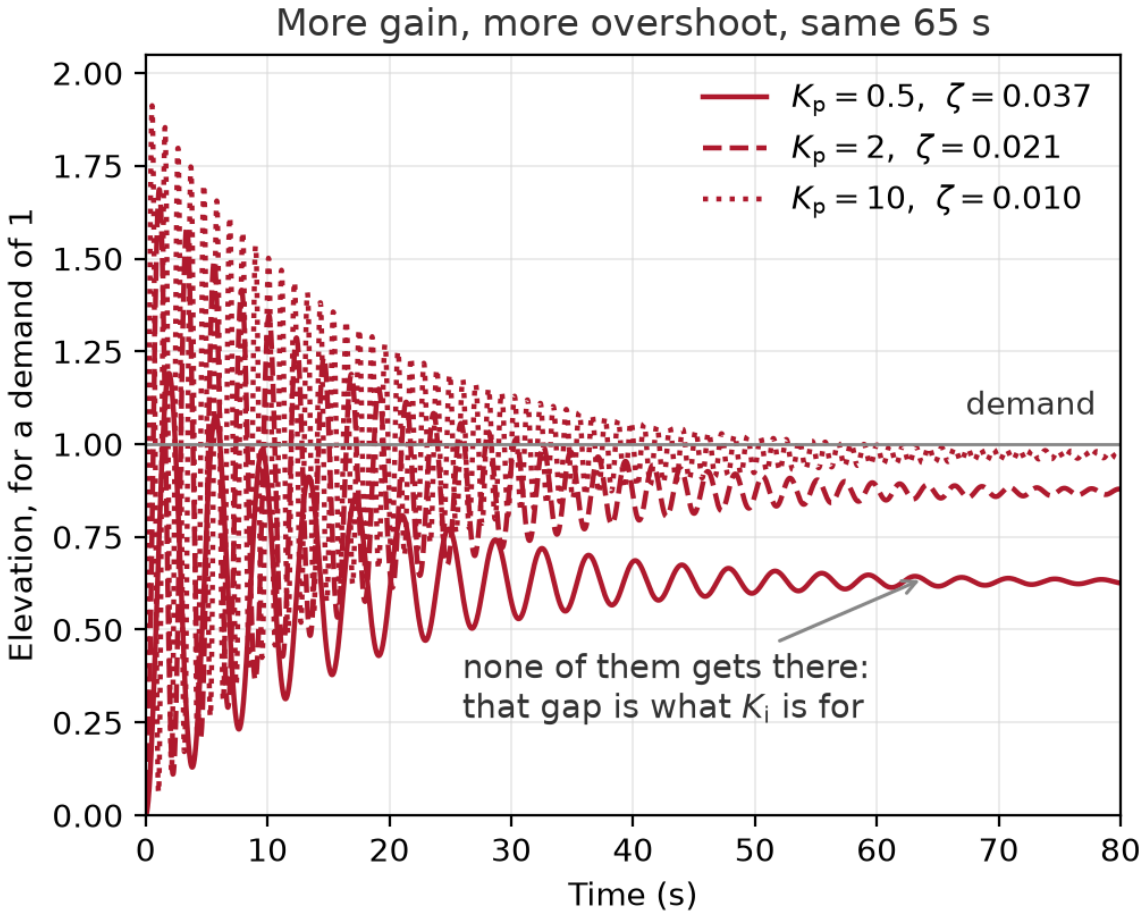
$$s^2 + 2\zeta\omega_n s + \omega_n^2(1 + KK_p)$$

**$K_p$ is not in the coefficient of s.**




The poles slide straight up
Imaginary
Real
Re = −ζω_n = −0.06,
whatever the gain
open loop, K_p = 0
6
4
2
0
−2
−4
−6
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10


More gain, more overshoot, same 65 s
Elevation, for a demand of 1
Time (s)
K_p = 0.5, ζ = 0.037
K_p = 2, ζ = 0.021
K_p = 10, ζ = 0.010
demand
none of them gets there:
that gap is what K_i is for
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
50
60
70
80

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# What counts as good?

## Agreeing it before we design

# A validatable requirement is a number with a test

| Quantity | Means | Typical |
|---|---|---|
| Overshoot $M_p$ | how far past it goes | "no more than 20%" |
| Settling time $t_s$ | until it stays in a band | "within 2% inside 8 s" |
| Steady-state error | what is left at the end | "under 1 degree" |

**A pilot's judgement gets you a long way. It cannot be met, missed, or worked with analytically.**

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# System identification

## Getting the model from the machine

YOU DO

# Three numbers, three measurements

- **Period** of the oscillation → damped frequency, $\omega_d = 2\pi/T_d$
- **Ratio of successive peaks** → damping ratio, by log decrement
- **Steady value** → the gain

> Do it **by hand first**. `tfest` fits any order in one line, and a fit you can understand and critique is more useful to you.

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# Design, and submit

Your gains, on the real machine

YOU DO

# Tune it, then send it

- Tune in simulation until it meets **the requirement we agreed**
- `s2_tune` **checks your gains** and gives you a pre-filled link to submit them to me
- You press Submit, so you see what goes out **under your name**

> Gains outside the envelope are **refused, not adjusted**. You will be told which limit you missed.

# We'll fly three rounds

1. **The extremes.** Most aggressive, most sluggish, most integral
2. **The cohort average.** Often worse than most of its parts
3. **The best few**

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# Simulation, meet hardware

## Where they disagree

# It met the spec. It still might not fly

- **The model is a simplification.** One axis, linearised, fitted over seconds
- **The actuators saturate.** Ask a 24 V amplifier for 30 V and it will not give it to you
- **There is noise.** The encoder quantises, and D amplifies exactly that
- **The plant moves.** Trim, friction, and the rig at 14:00 is not the rig at 13:00

**Each of these gets a week of its own. The answer is to design for it, not to model harder first.**

# Today, in five lines

- Control design is a **loop**, and every week is one step of it
- Feedback is a **tradeoff**: disturbance rejection against noise and instability
- A **requirement** is a number with a test, agreed *before* designing
- A model you **measured** beats a model you assumed, and knowing which you have matters
- **Simulation and hardware often disagree**, and we can design for that

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# There is no right answer until you say what you want

See you next lecture